Radiation Studies on the ATLAS Link and Control Daughterboard for the HL-LHC Upgrade

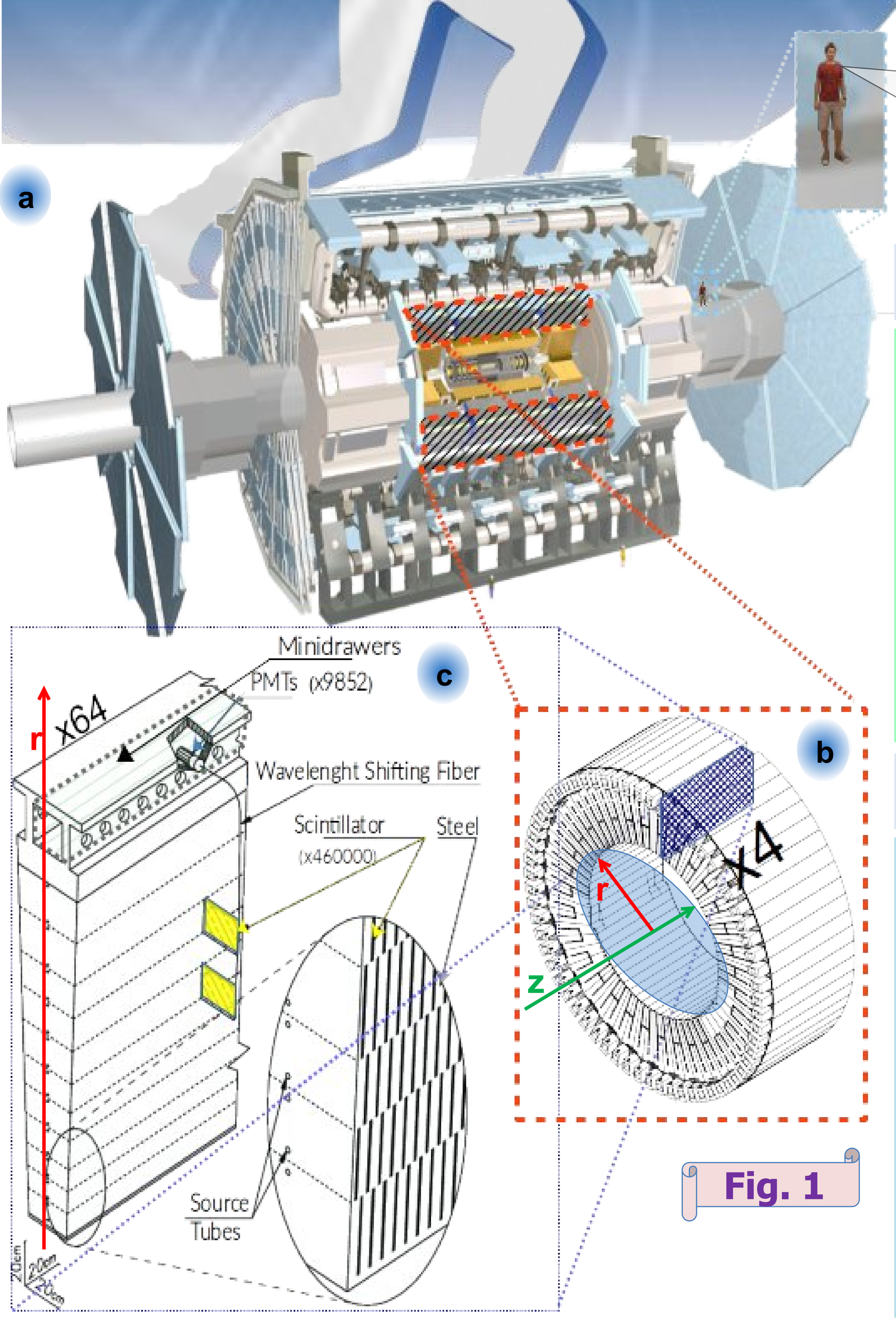

Fig. 1

The ATLAS Hadronic Tile Calorimeter (TileCal, Fig 1a)

- Sampling calorimeter to identify and measure transverse missing energy, hadronic jets, hadronically decaying tau-leptons and assists in muon identification[1].
- Functionally divided in four cylindrical barrels (Fig. 1b), each with 64 modules of steel absorber plates and plastic scintillator tiles (Fig. 1c).
- Scintillator light is collected by fibers and read by electronics interfaced with PMTs positioned in the girders (Fig. 1c).
- Electronics are being upgraded to handle higher radiation and data rates by providing continuous read-out of all TileCal with better timing and energy resolution.

High Luminosity Tilecal **on-detector** read-out system

The on-detector electronics are segmented in 896 **Minidrawers (MDs)** that continuously sample data from all TileCal PMTs at **40 MHz** (Fig. 2). Two PMTs/cell for redundancy. Each **MD** hosts:

- **12 PMTs** powered by **High Voltage Active Dividers (HVAD)**
- **12 Front-End Boards (FEBs)** to shape and condition the PMT signals,
- **a Mainboard (MB)** to provide power received from the **Low Voltage Power Supply (LVPS)** system and continuously sample and digitize **two gains** of shaped PMT signals,
- **a Daughterboard (DB,** Fig. 3) to receive and distribute **LHC synchronized timing**, configuration and control to the front-end, and to send **continuous read-out** of the digital data from all the MB channels to the off-detector systems via multi-Gbps optical links

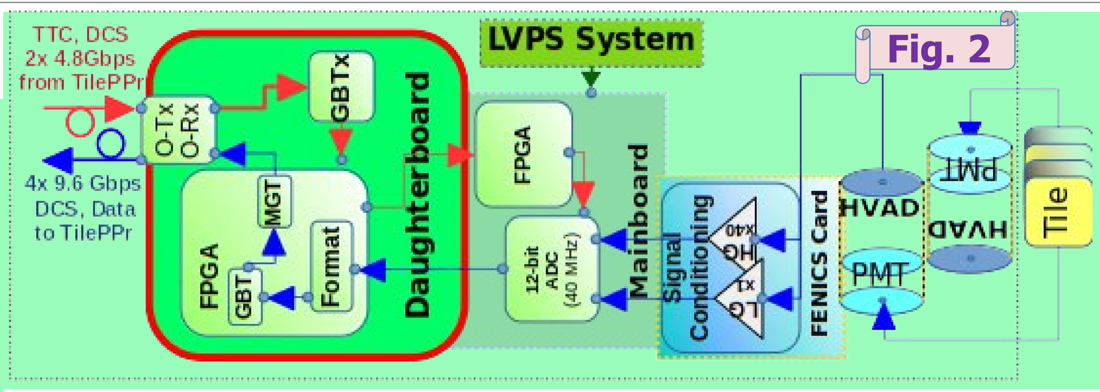

Fig. 2

The Daughterboard revision 6

- A radiation tolerant board with two functionally-equivalent halves (A and B)[2]. On each DB half:
  - An **SFP+** RX channel and a **CERN rad-hard GBTx ASIC** [3] receive a **4.8 Gbps GBT-FEC downlink** with commands and clocks from the off-detector
  - A **Xilinx Kintex Ultrascale (KU) FPGA** distributes clocks and configuration received from the GBTx to the MB, while reading slow integrator data and fast digitized data from two different gains of the PMT channels.
  - The xADCs of the KU FPGAs sample temperature, voltage stability, currents.
  - The KU FPGA sends of two copies of GBT-FEC protected words off-detector through two 9.6 Gbps uplinks via two **SFP+** TX.
  - The production of the boards will take place starting on 2026.

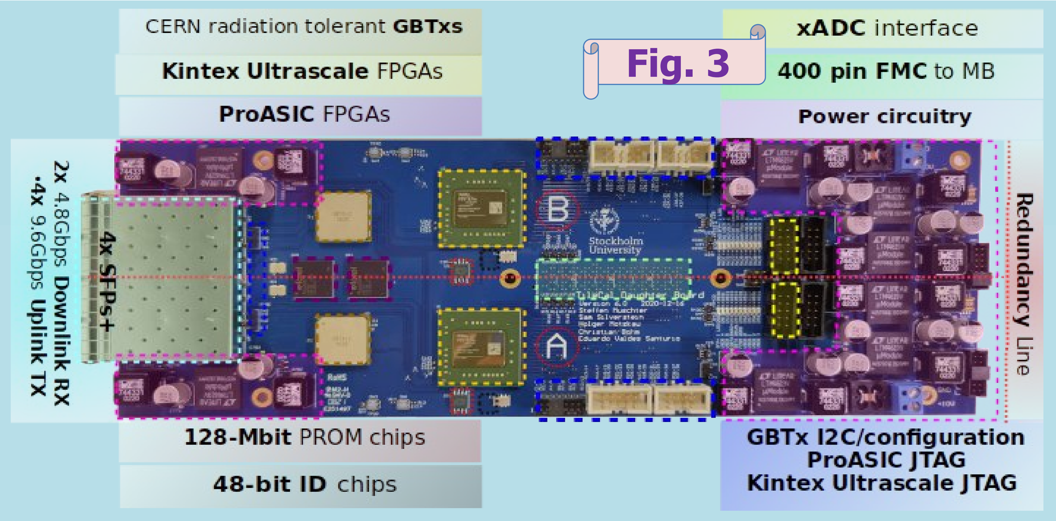

Fig. 3

- Radiation tolerance was engineered into the design by adding **redundancy layers**, using the **Xilinx Soft Error Management (SEM), Triple Mode Redundancy (TMR), GBT-FEC** protocol, and **GBTx** ASICs.

Radiation requirements for DB Production

- The radiation requirements for Tota Ionizing Dose (**TID**), Non-Ionizing Energy Losses (**NIEL**) and Single Event Effects (**SEE**) were established by the voxel with the highest simulated irradiation value, affecting 128 out of the 896 DB positions (Fig. 4a, 4b and 4c).
- During production phase, 10 samples of each batch of the active components acquired were tested to account for possible batch differences (except for big/highly dense components like FPGAs)
- **Safety Factors** were applied to account for inconsistencies in **simulation (1.5)** and in tests with **mono-energetic beams (1.3)**
- The requirements of: **TID: 32.71 Gy, NIEL: 4,38x10E12 (1 MeV n)/cm2, and SEE: 2.37x10E11 h/cm2.**
- The components were pre-selected and tested with higher dose and fluence to account for lot variation [4, 5].

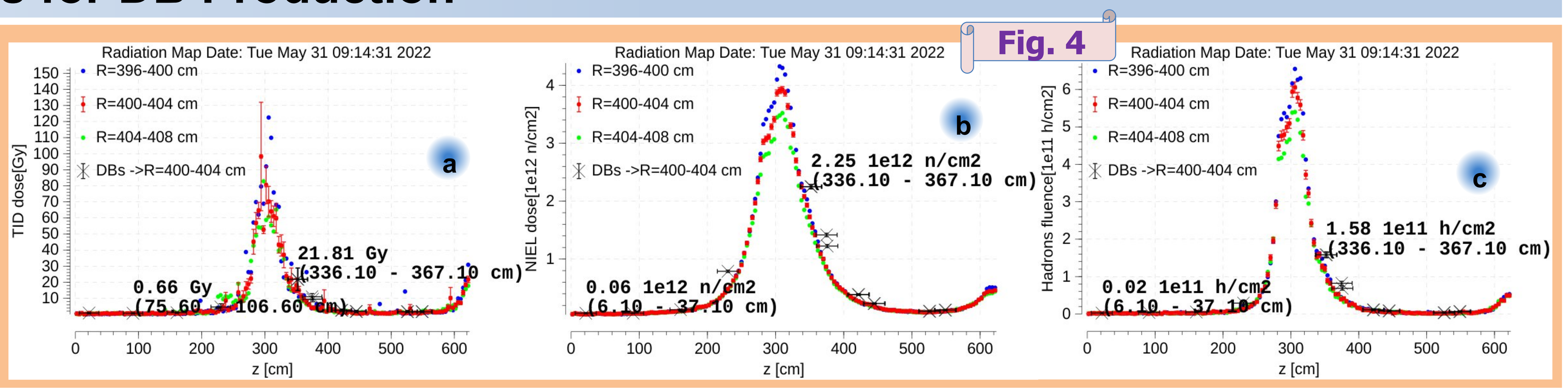

Fig. 4

Radiation tests for production components

- **TID** to **50 Gy @ 3.24 Gy/h** (Fig. 5) and **NIEL to 5x10E12 (1 MeV n)/cm2** tests were done in 60 samples (6 lots) of MOSFETs (FDFMA3N109). The 50 Gy TID exposure reduced the voltage threshold of the MOSFETs by around 15 mV. The NIEL exposure did not affect the functionalities of the irradiated devices.
- **TID** to **50 Gy @ 3.24 Gy/h** (Fig. 6a and 6b) and **NIEL** tests were done in 20 samples (2 lots) of Instrumentation amplifiers (INA333) configured with gain ~500. Linearity and Gain stability were stable over the TID tests and verified after the NIEL exposure. The performance of the devices was not affected by the radiation.
- **TID** and **NIEL** tests were done in 10 Direktronik SFP+ samples (batch 20122113 of model 33-4248) with no impact on the performance of the samples
- **SEE** tests were performed in 3 Direktronik SFP+ samples that integrated: **TID: 167 Gy, NIEL: 2.61x10E12 (1 MeV n)/cm2,** and **High Energy Hadrons fluence: 7.78x10E11**
  - **TID** effects and **SEL** occurrences were not detected during the current monitoring of the devices over the irradiation period. The currents were stable and the devices functioned well over 12 power cycles (Fig. 7a).
  - **SEU** rates were studied over 8 runs (Fig. 7b and 7c), where average rates between **1x10xE-8 errors/(h cm2)** and **8x10xE-8 errors/(h cm2)** were measured over all the runs for the three irradiated modules
    - Accounting for the average rate of **5.67x10E-8 errors/(h cm2)** (worst performing module), a **total of 13440 errors per SFP+** can be expected over 10 years of HL-LHC
    - **9.15 bit errors/min** are expected in all TileCal, which will be mitigated by the **GBTx GBT-FEC** protocol capable of handling 16 consecutive errors every 25 ns.
  - Two rare occurrences of what seems to be **SEFI-like** behaviour happened during the runs and were recovered via power cycling
    - The rate of the SEFI-like occurrence is **0.65 SEFI/day** for a total of **2389 SEFI** over the whole HL-LHC period
    - These events will be further studied, but wont impact the data taking since can be easily managed by power cycles, resets and the redundancy layers.
- **TID** to **50 Gy @ 3.42 Gy/h** (Fig. 8a and 8b) and **NIEL to 5x10E12 (1 MeV n)/cm2** tests were done in 10 samples (1 lot) of FLASH memories (IS25LP256). The exposure did not impact the behaviour and current consumption of any of the functionalities of any of the irradiated devices. The NIEL exposure did not affect the functionalities of the irradiated devices.
- **TID** to **50 Gy @ 3.24 Gy/h** (Fig. 9a and 9b) and **NIEL to 5x10E12 (1 MeV n)/cm2** tests were done in 40 samples (4 lots) of 40 MHz and 100 MHz oscillators. The current consumption and the frequencies of all the devices were stable over the whole TID exposure. The NIEL exposure did not affect the functionalities of the irradiated devices.

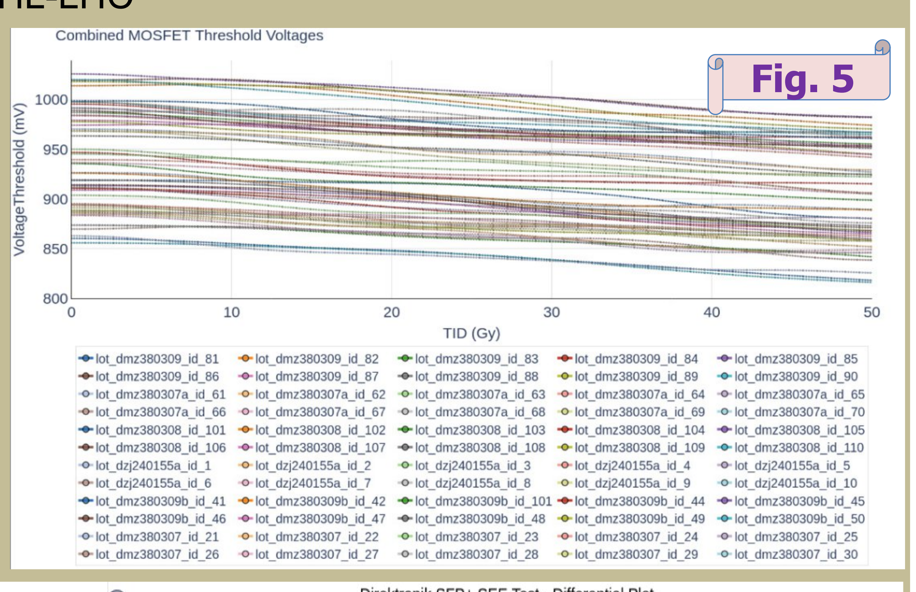

Fig. 5

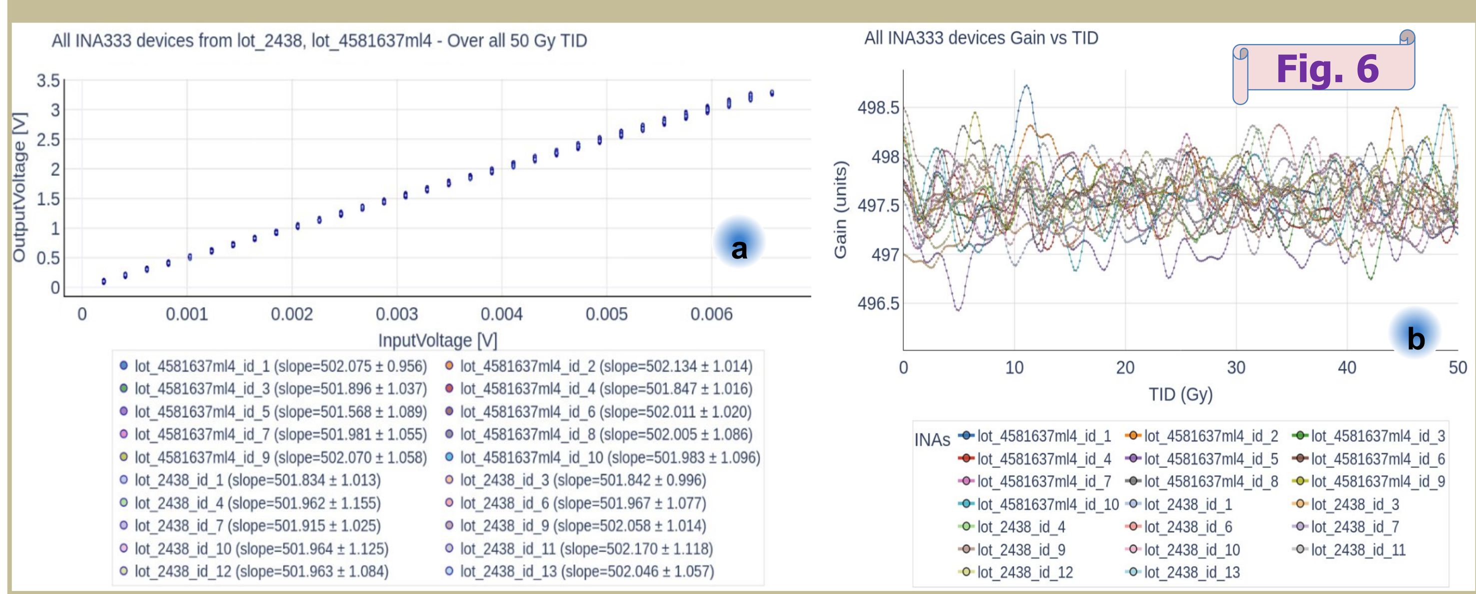

Fig. 6

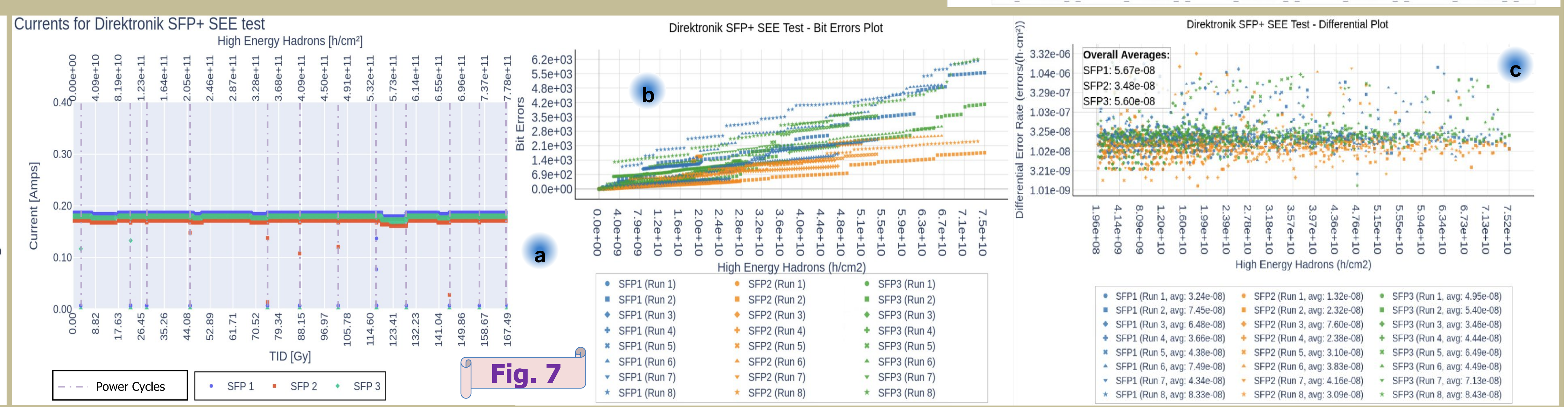

Fig. 7

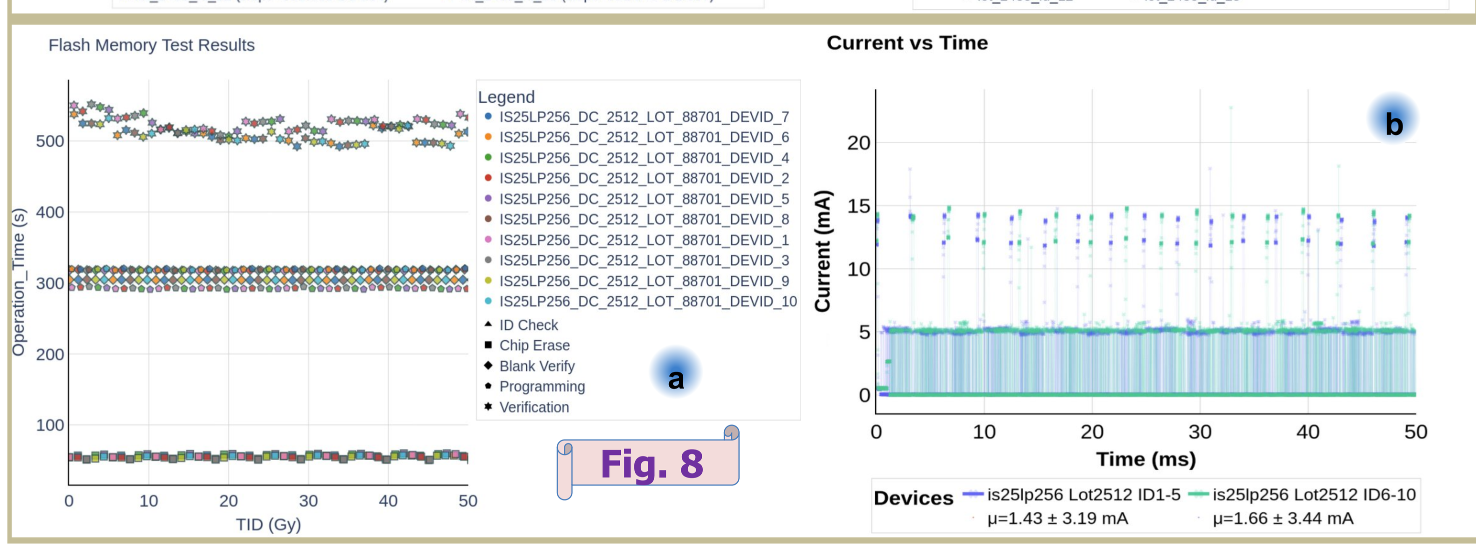

Fig. 8

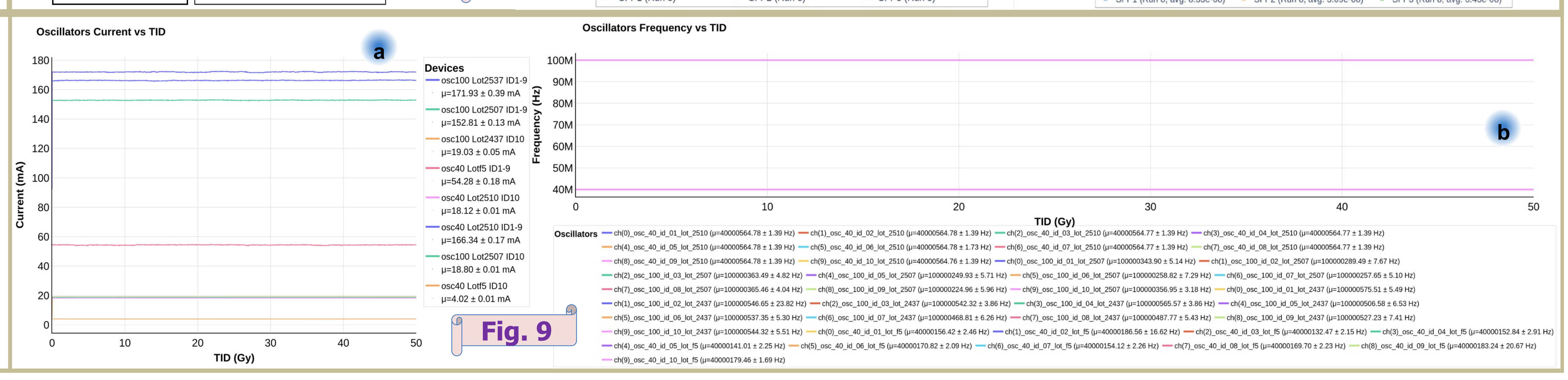

Fig. 9

Conclusions

The component selection tested for the production of the ATLAS TileCal DB fulfill the HL-LHC radiation requirements following the current collaboration guidelines and simulations for TID, NIEL and SEE. Most of the components have been acquired and qualified in radiation on a lot basis for pre-production and production phases. Radiation campaigns to test the DC-DC converters (LTM4691), ProASIC and KU FPGA acquired lots will take place during end 2025 and 2026. Around 930 DBs will be produced during 2026 and 2027 as the contribution of Stockholm University to the HL-LHC era for TileCal.

Relevant references

[1] ATLAS collaboration. Technical Design Report for the Phase-II Upgrade of the ATLAS Tile Calorimeter. CERN-LHCC- 234 2017-019, ATLAS-TDR-028, 2018

[2]] E. Valdes Santurio, Development of the read-out link and control board for the ATLAS Tile Calorimeter Upgrade, Ph.D. Thesis, Stockholm University (2019) [ISBN: 978-91-7797-896-1]

[3] K. Wyllie, P. Moreira and J. Christiansen, GBTX MANUAL V0.15 Technical Report, (2021),

[4] E. Valdes, Radiation studies performed on the High Luminosity ATLAS TileCal link Daughterboard, DOI: 10.1088/1748-0221/18/04/C04011

[5] E. Valdes , A radiation tolerant read-out link and control Daughterboard for the upgrade of the ATLAS Hadronic Calorimeter, ATL-TILECAL-SLIDE-2023-619, https://cds.cern.ch/record/2880342/files/ATL-TILECAL-SLIDE-2023-619.pdf

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